PoissonNLLLoss#

https://pytorch.org/docs/stable/generated/torch.nn.modules.loss.PoissonNLLLo

Description:

Negative log likelihood loss with Poisson distribution of target. The loss can be described as: The last term can be omitted or approximated with Stirling formula. The approximation is used for target values more than 1. For targets less or equal to 1 zeros are added to the loss. log_input (bool, optional) – if True the loss is computed as exp(input) −target∗input\exp(\text{input}) - \text{target}*\text{input}exp(input) −target∗input, if False the loss is input−target∗log(input+eps)\text{input} - \text{target}*\log(\text{input}+\text{eps})input−target∗log(input+eps). full (bool, optional) –

Function Signature

```
class torch.nn.modules.loss.PoissonNLLLoss(log_input=True,
full=False, size_average=None, eps=1e-08, reduce=None,
reduction='mean')[source]#

Parameters

Shape:

forward(log_input, target)[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> loss = nn.PoissonNLLLoss()
>>> log_input = torch.randn(5, 2, requires_grad=True)
>>> target = torch.randn(5, 2)
>>> output = loss(log_input, target)
>>> output.backward()
```

Detailed Documentation

Documentation

Negative log likelihood loss with Poisson distribution of target.

The loss can be described as:

The last term can be omitted or approximated with Stirling formula. The approximation is used for target values more than 1. For targets less or equal to 1 zeros are added to the loss.

log_input (bool, optional) – if True the loss is computed as exp(input) −target∗input\exp(\text{input}) - \text{target}*\text{input}exp(input)−target∗input, if False the loss is input−target∗log(input+eps)\text{input} - \text{target}*\log(\text{input}+\text{eps})input−target∗log(input+eps).

full (bool, optional) –

whether to compute full loss, i. e. to add the Stirling approximation term

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True

eps (float, optional) – Small value to avoid evaluation of log(0)\log(0)log(0) when log_input = False. Default: 1e-8

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: $(*)$(*)$(*)$, where $*$*$*$ means any number of dimensions.

Target: $(*)$(*)$(*)$, same shape as the input.

Output: scalar by default. If reduction is 'none', then $(*)$(*)$(*)$, the same shape as the input.

Runs the forward pass.